An Explainable NLP Framework for Suicide Risk Classification, Evidence Extraction, and Factor Identification from Social-Media Text

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***Abstract*—Suicide-risk assessment from social-media text needs reliable severity classification and an account of why a post received its label. We address the IEEE BigData 2026 Explainable Suicide Risk Detection Challenge, which asks for (i) a four-level risk label (Indicator, Ideation, Behavior, Attempt) with verbatim supporting evidence, and (ii) multi-label identification of suicide-related factors from a 24-category taxonomy. We present a lightweight, fully inspectable pipeline: word and character TF-IDF with logistic regression for risk, a phrase-level span scorer that returns exact substrings of the post as evidence, and a one-vs-rest logistic regression with per-factor decision thresholds for factors. All splits are grouped by user to prevent leakage, and design decisions are checked across multiple split seeds, which led us to reject several changes that looked beneficial on a single split. On a held-out, user-disjoint validation split (380 posts, 31 users), the system obtains a risk Weighted F1 of 0.6468 (5-seed mean 0.6356), a local-approximation Phrase F1 of 0.5599 (3-seed mean 0.5246), and a factor Macro F1 of 0.4419 (3-seed mean 0.4235). Evidence quality is concentrated in the Indicator class; for the other three classes the local Phrase F1 is 0.322. These are local validation results, not official leaderboard scores. Transformer components are implemented but have no completed evaluation, and none is reported.**

***Index Terms*—Suicide risk detection, natural language processing, explainable AI, evidence extraction, multi-label classification, TF-IDF, group-wise validation, mental-health NLP**

I. Introduction

A. Background and Research Problem

People in distress often describe suicidal thoughts in online communities, and automated analysis of such text has been studied as a screening and triage aid [2], [3]. The task is technically hard for several reasons. Severity is expressed indirectly and depends on context, negation, and intent; adjacent severity levels share much of their vocabulary; social-media text is informal and noisy; the label distribution is imbalanced; and corpora typically contain several posts per author, so that naive random splitting leaks author-specific style into evaluation.

Ordinary text classification is insufficient in this setting. A bare label offers a human reviewer no basis for checking or contesting the output, which is problematic in a high-stakes domain where interpretable models are preferable to opaque ones when they are competitive [4]. The challenge we address therefore couples classification with two explanatory outputs: the exact phrases that support the risk level and the psychosocial factors present in the post.

B. Challenge Context and Research Gap

The IEEE BigData 2026 Explainable Suicide Risk Detection Challenge [1] provides Reddit r/SuicideWatch posts annotated with a risk level, supporting evidence phrases, and factor labels. Handling severity classification, span-level evidence, and 24-way multi-label factors *jointly*, on a small, noisy, class-imbalanced corpus with user-level dependency, is the gap this work addresses. Most prior suicide-risk work targets label prediction only (Section III).

C. Contributions

Each contribution below is implemented and evaluated in our notebook.

1) A three-component pipeline for four-class risk classification, verbatim evidence extraction, and 24-category multi-label factor identification, producing one explainable output per post.
2) A phrase-level evidence extractor over sentence and subclause candidates whose outputs are exact substrings of the source post by construction.
3) Word (1–2-gram) plus character (3–5-gram) TF-IDF features for risk, and per-factor decision thresholds for factors.
4) A user-grouped, multi-seed validation protocol, used to accept or reject every modeling change.
5) A documented set of rejected techniques (larger C, SVM/LR ensembling, character features for factors, oversampling, greedy span shrinking, CRF tagging) with the evidence for rejecting each.

II. Problem Formulation

Let x be a post. We learn three mappings. The risk model $f_r(x) \rightarrow y_r \in \{\text{Indicator}, \text{Ideation}, \text{Behavior}, \text{Attempt}\}$; the evidence model $f_e(x) \rightarrow E = \{e_1, \ldots, e_m\}$, where every e_i is a contiguous substring of x ($E = \emptyset$ is allowed and, for Indicator posts, is the usual annotation, written "none"); and the factor model $f_f(x) \rightarrow Y_f \subseteq \mathcal{F}$, $|\mathcal{F}| = 24$.

Weighted F1 (risk) averages class F1 by class support n_c among N posts:

$$F1_w = \sum_{c=1}^{4} \frac{n_c}{N} F1_c. \tag{1}$$

Macro F1 (factors) averages the F1 of the $K{=}24$ binary factor decisions equally, so rare factors weigh as much as common ones:

$$F1_{\text{macro}} = \frac{1}{K}\sum_{k=1}^{K} F1_k. \tag{2}$$

Phrase F1 (evidence). Our local evaluator implements the span-matching rule as documented in the challenge material available to us: a predicted span p matches a reference span g if, ignoring case, one contains the other and $|\text{tok}(p)| \leq 3\,|\text{tok}(g)|$; matching is greedy and one-to-one. For post i with m_i matches, $\hat{E}_i$ predicted and E_i reference spans, $P_i = m_i/|\hat{E}_i|$, $R_i = m_i/|E_i|$, $F_i = 2P_iR_i/(P_i + R_i)$; a post with no reference and no prediction scores 1, and one with exactly one of the two empty scores 0. Phrase F1 is the mean of F_i. Because we do not have the organizers' scoring code, we call this a *local approximation*, and all scores below are validation scores rather than official results.

III. Related Work

Mental-health NLP and suicide-risk detection. Early work showed that language on social media carries signal about depression [5] and suicide risk [2]; Ji *et al.* survey the ML methods used for suicidal-ideation detection [3]. Severity-level formulations include the CLPsych 2019 shared task on Reddit posts [6], built on expert and crowdsourced annotations [7], and knowledge-aware severity assessment aligned with a clinical rating scale [8]. These studies predict labels; none of them requires span-level evidence, which is the distinguishing element of the present challenge.

Transformers. BERT [9] and its distilled variant DistilBERT [10] are standard baselines, and domain-adaptive models pretrained on mental-health text, such as MentalBERT [11], can improve over general models. Their drawbacks for our setting are compute cost, opacity, and sensitivity to small training sets. We implemented a DistilBERT pathway but have no completed evaluation (Section VI-D).

Linear models and TF-IDF. TF-IDF weighting [12] with regularized linear classifiers [13] remains a strong, inspectable baseline on small corpora, and character-level features are known to help on noisy text [14]. We use scikit-learn implementations [15].

Explainability and evidence. Post-hoc explainers such as LIME [16] and SHAP [17] attribute importance to features, while rationale benchmarks such as ERASER [18] evaluate whether extracted text supports a prediction. Our evidence extractor is closer to the rationale view: it selects supporting text directly rather than explaining a separate model.

Multi-label learning and thresholds. One-vs-rest decomposition is a standard multi-label baseline [19], and per-class thresholds can improve F1 under imbalance [20]. Group-wise validation is needed when samples are not independent [21].

IV. Dataset and Data Analysis

A. Data and Annotations

The training file contains 1,635 labeled posts (columns: `row_id`, `anon_user_id`, `post_id`, `post`, risk label, evidence, factors) from 153 anonymized users (7–30 posts per user; mean 10.7, median 9). The prediction file contains 502 unlabeled posts from 48 users, none of whom appears in the training data (verified in our notebook). Evidence is a semicolon-separated list of phrases; factors are stored as a string-encoded list.

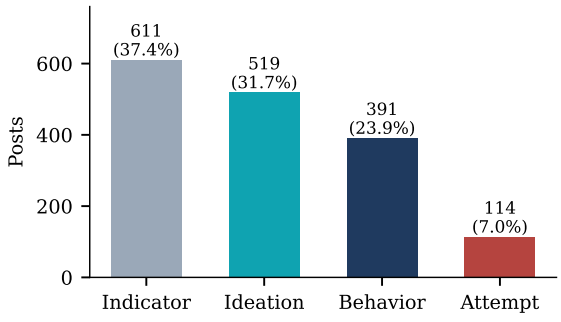


Fig. 1. Risk-level distribution of the 1,635 training posts.

B. Risk Labels and Class Imbalance

After case and whitespace normalization of ten raw spellings, the labels are Indicator 611 (37.4%), Ideation 519 (31.7%), Behavior 391 (23.9%), and Attempt 114 (7.0%) (Fig. 1). Attempt is thus about one-sixth as frequent as Indicator.

C. Evidence Characteristics

Of the training posts, 583 (35.7%; 95.4% of Indicator posts) are annotated "none", 5 Indicator posts have a missing evidence field, and all remaining posts carry at least one phrase. The 1,047 posts with evidence contain 1,833 phrases (1.75 per post on average) with a median length of 4 whitespace tokens (90th percentile 8). Evidence annotations are therefore short, and their presence is itself strongly tied to the risk class.

D. Factors

The taxonomy has 24 categories, in a long-tailed distribution (Fig. 2): hopelessness occurs in 745 posts, whereas sexual-orientation-related issues occurs in 8. Factor lists repeat labels in 812 posts, so we deduplicate within each post; 128 posts have no factor and the average is 2.92 unique factors per post.

E. Text Length and Data Quality

Posts are short and heavy-tailed: median 38 tokens, mean 68.8, maximum 2,066. Median length grows with severity (Indicator 20, Ideation 45, Behavior 45, Attempt 78 tokens). Quality issues are limited: 47 rows repeat the text of another row (1,588 unique texts), 10 raw risk spellings collapse to 4 classes, and no post is empty after cleaning.

V. Preprocessing and Validation Strategy

A. Minimal, Non-Destructive Preprocessing

We convert missing posts to empty strings, collapse repeated whitespace, and lowercase text for the TF-IDF models. We

TABLE I
COMPARISON WITH REPRESENTATIVE PRIOR WORK

Study	Task	Method	Strength	Limitation	Relevance
Coppersmith [2]	Screening for suicide risk from social media	Neural text classifiers	Demonstrates feasibility	Label only; no evidence	Motivation
Zirikly [6]	Four-level risk, Reddit	Shared-task systems (various)	Public benchmark, expert labels	Label only	Closest task framing
Gaur [8]	Severity aligned with a clinical scale	Knowledge-aware models	Clinically grounded labels	No span evidence	Severity taxonomy
Ji [11]	Mental-health language modeling	Domain-adaptive pretraining	Strong text representations	Compute; opaque	Future work
DeYoung [18]	Rationalized NLP	Rationale benchmark	Evaluates evidence faithfulness	Not mental-health specific	Evidence concept
This work	Risk + evidence + 24 factors	TF-IDF + linear models, user-grouped CV	Inspectable; verbatim evidence	Small data; classical ceiling	—

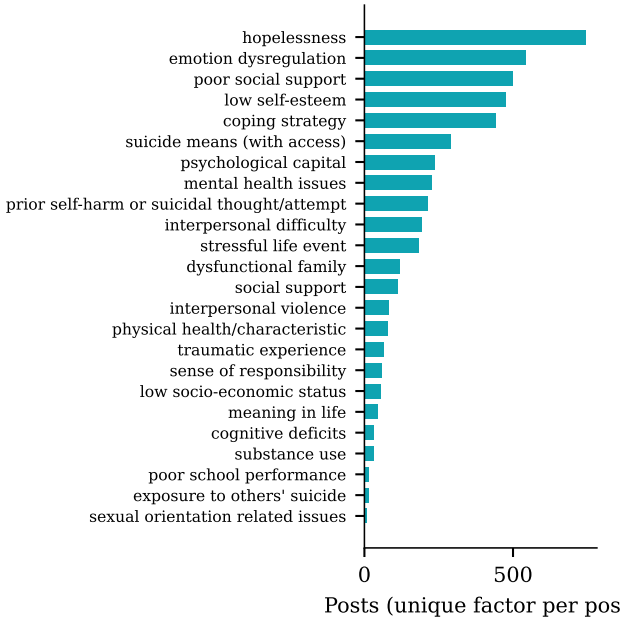


Fig. 2. Frequency of the 24 factor categories in the training data (unique factor per post).

deliberately do *not* remove punctuation, stop words, or negations: tokens such as "not", "never", and "don't" can reverse the meaning of a statement in this domain. Original-case text is preserved for transformer use and for evidence output.

B. User-Grouped Splitting

Users contribute many posts, and a user's phrasing is stylistically consistent, so row-level splits would let the model memorize authors. All splits use scikit-learn's `GroupShuffleSplit` on `anon_user_id` with a 20% validation share. The primary split (seed 42) yields 1,255 training posts (122 users) and 380 validation posts (31 users) with zero user overlap, mirroring the user-disjoint design of the prediction file.

C. Multi-Seed Validation

A single 380-post split gives noisy estimates: the risk Weighted F1 of the same baseline ranges from 0.588 to 0.651 across five splits. We therefore re-evaluate each candidate on seeds $\{42, 52, 62, 72, 82\}$ for risk, and on seeds $\{42, 52, 62\}$ for evidence and factors, and accept a change only if its mean improves. All seeds are drawn from the labeled training file; the final models are fit on the primary split's training portion (1,255 posts) rather than refit on all labeled data.

VI. PROPOSED METHODOLOGY

Fig. 3 summarizes the system. Hyperparameters not listed are scikit-learn defaults.

A. Risk Classification

Each lowercased post is mapped to a word TF-IDF vector (1–2-grams, `min_df`=2, at most 10,000 features) concatenated with a character TF-IDF vector (3–5-grams, `min_df`=2, at most 10,000 features). A multinomial logistic regression with C=1, `class_weight=balanced`, and `max_iter`=1000 predicts the four classes. Character n-grams give partial-word coverage of misspellings, elongations, and informal spellings that exact word matching misses, and are cheap to add. Word-only features (Weighted F1 0.6442 on the primary split) serve as the baseline.

B. Evidence Extraction

Evidence extraction (Fig. 4) is framed as binary classification of candidate spans. Each post is split into sentences (at line breaks and sentence-final punctuation) and each sentence into sub-clauses at semicolons, colons, commas, and the conjunctions *and, but, or, because, so, since, although, though, when, while*; candidates are deduplicated. On the primary training split this yields 19,045 candidates, of which 1,608 (8.4%) are positive under the local matching rule of Section II. A TF-IDF (1–2-gram, `min_df`=2, 5,000 features) logistic regression with balanced class weights scores each candidate. Candidates with probability $\geq t$ are sorted by token length, and a candidate is dropped if it contains a shorter kept span, which prefers the shortest valid evidence. The threshold $t = 0.7$ was chosen from $\{0.2, \ldots, 0.8\}$.

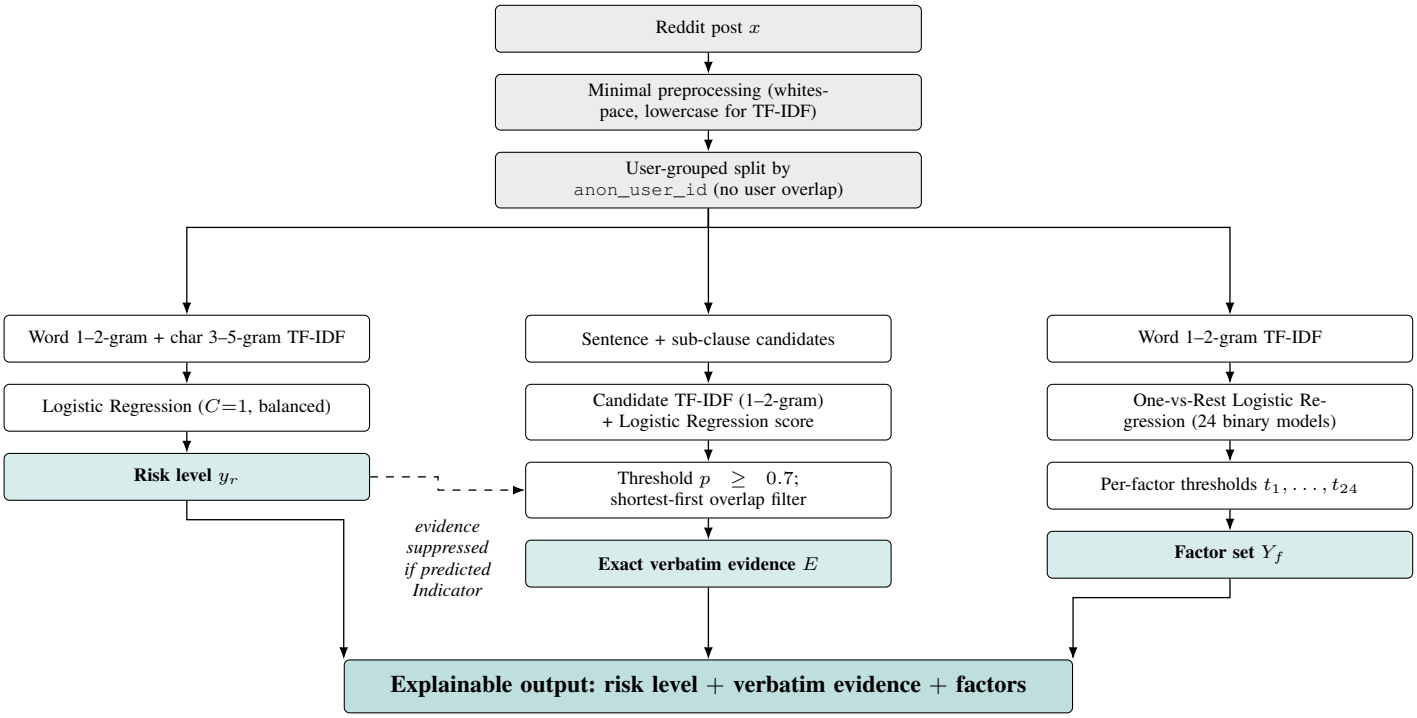


Fig. 3. Overall architecture of the proposed explainable framework. The three branches share one user-grouped split; the final submission suppresses evidence for posts predicted as Indicator.

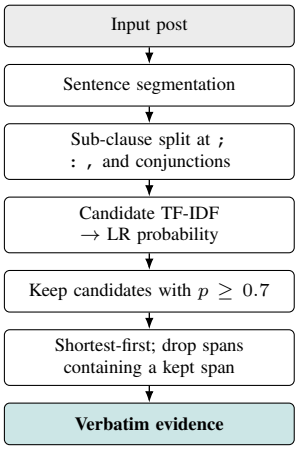


Fig. 4. Evidence-extraction pipeline. No text is generated or paraphrased.

The system does not generate or paraphrase evidence: every output is a substring of the original post, which makes each explanation checkable against its source. In the final submission, spans longer than 80 tokens are discarded and evidence is left empty for posts predicted as Indicator, matching the annotation convention for that class. We verified that all 377 spans in the submitted prediction file (206 posts) are exact substrings of the original post text, and that re-executing the notebook regenerates that file byte for byte.

C. Factor Identification

Factors are predicted by one-vs-rest logistic regression (C=1, balanced) on word TF-IDF (1–2-grams), giving 24 independent binary decision functions $p_k(x)$. A single cutoff of 0.5 is poorly matched to factors whose frequencies differ by two orders of magnitude, so each factor has its own threshold,

$$t_k = \arg \max_{t \in \{0.20, 0.25, \ldots, 0.60\}} F1_k(t), \qquad \hat{y}_k = \mathbb{1}[p_k(x) \geq t_k], \tag{3}$$

selected on the validation split. Note that the reported tuned scores are therefore optimistic by construction; we quantify this with multi-seed means.

D. Transformer Extension

The notebook contains a DistilBERT [10] risk classifier (`distilbert-base-uncased`, maximum length 256, batch size 8, learning-rate candidates $\{10^{-5}, 2\times10^{-5}, 5\times10^{-5}\}$, up to 6 epochs, weight decay 0.01, early-stopping patience 2, class-weighted cross-entropy), a multi-label counterpart for factors, and an ensemble search. These were implemented and made execution-ready, *but no validated Transformer result is reported*: the notebook records only an incomplete training log for the first learning rate, with no finished sweep, selected checkpoint, or test-time use, and its model-selection table marks Transformer and ensemble as not run. Selection of a Transformer by the same validation split would also need the multi-seed treatment applied to the classical models.

TABLE II
LOCAL VALIDATION PERFORMANCE OF THE FINAL MODELS

Component	Metric	Primary split	Multi-seed mean (std)	Seeds
Risk	Weighted F1	0.6468	0.6356 (0.0221)	5
Evidence	Phrase F1†	0.5599	0.5246 (0.0272)	3
Factors	Macro F1	0.4419	0.4235 (0.0179)	3

†Local approximation of the official metric. Thresholds (t=0.7, t_k) were tuned on the primary split, so its scores are optimistic. These are validation scores, not leaderboard scores.

TABLE III
RISK CLASSIFICATION: PER-CLASS RESULTS (PRIMARY SPLIT)

Class	Precision	Recall	F1	Support
Indicator	0.786	0.815	0.800	162
Ideation	0.566	0.571	0.569	105
Behavior	0.530	0.530	0.530	83
Attempt	0.478	0.367	0.415	30

VII. EXPERIMENTAL SETUP

All experiments use Python with scikit-learn [15]. Classical models run on a CPU in seconds to minutes; we report no training times. The random state is 42 unless a seed is stated. We report precision, recall, and F1 only where the notebook computes them.

VIII. RESULTS AND ANALYSIS

A. Main Results

Table II gives the headline results. For risk, accuracy on the primary split is 0.6500 and macro-averaged F1 is 0.58 (Table III). The primary split is a favorable draw for all three components: its score exceeds the multi-seed mean each time, so the means are the more conservative estimates.

B. Model Comparison

Table IV compares the executed configurations. Linear SVM and the LR+SVM blend do not beat plain logistic regression on the multi-seed mean, and the blend has the larger variance (std 0.0301 vs. 0.0221).

C. Ablation and Robustness

Character features raise the risk Weighted F1 on all five seeds (Fig. 5, Table V), by $+0.0141$ on average (the gain per seed ranges from $+0.0026$ to $+0.0245$). We did not run a significance test, and five overlapping draws from 153 users are too few to support one.

Table VI lists accepted and rejected techniques. Two cases illustrate why multi-seed checking matters: C=2 and greedy span shrinking each looked best on the primary split but did not survive averaging. The per-factor threshold gain is the largest single effect for factors, and it holds on each of the three seeds (+0.0562, +0.0489, +0.0361); it is, however, partly an artifact of selecting thresholds on the evaluation split. Two structurally different evidence methods (span classification and CRF tagging) reach similar means near 0.52, which suggests a ceiling for classical methods at this data size.

TABLE IV
MODEL COMPARISON (EXECUTED MODELS ONLY UNLESS STATED)

Task	Model / features	Score	Status
Risk	LR, word+char	0.6356[a]	Selected
Risk	LR, word only	0.6215[a]	Baseline
Risk	LR, word+char, C=2	0.6290[a]	Rejected
Risk	Linear SVM, word+char	0.6266[a]	Rejected
Risk	LR+SVM ensemble	0.6308[a]	Rejected
Risk	DistilBERT	—	Not evaluated
Evidence	Phrase-level LR, t=0.7	0.5246[b]	Selected
Evidence	Sentence-level LR	0.4799[c]	Baseline
Factors	OvR LR, word, per-factor t_k	0.4235[b]	Selected
Factors	OvR LR, word, t=0.5	0.3763[b]	Baseline
Factors	DistilBERT multi-label	—	Not evaluated

[a]Weighted F1, 5-seed mean. [b]3-seed mean. [c]Primary split only. All validation, user-grouped.

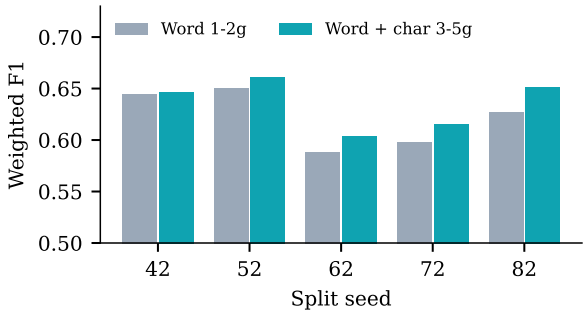


Fig. 5. Risk Weighted F1 on five user-grouped splits. Adding character 3–5-grams improves every seed (axis begins at 0.5).

D. Confusion Matrix and Risk Error Analysis

Fig. 6 shows that errors concentrate between neighboring severity levels. Indicator–Ideation confusion runs both ways (16 Indicator and 22 Ideation posts misclassified into the other), 21 Ideation posts are labeled Behavior, and 18 Behavior posts Ideation, while 9 Behavior posts are labeled Attempt. Attempt is the weakest class: only 11 of 30 are recovered, and 12 are predicted as Ideation, a non-adjacent error. Severity categories are ordinal, differ by tense and degree of action rather than by topic, and are therefore only partly separable by lexical features; the small Attempt class (114 training posts) compounds this. We do not draw conclusions beyond these counts, since 30 Attempt posts are a small sample.

IX. EXPLAINABILITY AND EVIDENCE ANALYSIS

Three-level explanation. For each post the system returns (1) a risk level, (2) verbatim supporting text, and (3) the factor categories present, e.g., *Risk: Ideation; Evidence: "wanna die"; Factors: hopelessness* for an annotated training example in which a user writes that the future will never improve. This structure lets a reviewer verify the evidence against the source, notice when it is absent or implausible, and see which factors co-occur with the label. We describe this output as *model-*

TABLE V
RISK WEIGHTED F1 ACROSS SPLIT SEEDS

Features	42	52	62	72	82	Mean	Decision
Word	.6442	.6505	.5879	.5979	.6269	.6215	Baseline
Word+char	.6468	.6610	.6040	.6149	.6514	.6356	Adopted

TABLE VI
ABLATION AND REJECTED TECHNIQUES

Technique	Result	Decision
Char 3–5g (risk)	+0.0141 mean, 5/5 seeds	Adopted
LR C=2 (risk)	Best on seed 42 (0.6519); 5-seed mean 0.6290 $<$ 0.6356	Rejected
LR+SVM ensemble	Mean 0.6308 $<$ 0.6356; higher std	Rejected
Char 3–5g (factors)	Macro F1 0.3760 vs. 0.3857	Rejected
Phrase-level vs. sentence candidates	0.5599 vs. 0.4799 (primary); thresholds 0.5/0.6/0.7: 0.5035/0.5166/0.5246 (3-seed mean)	Adopted (t=0.7)
Per-factor thresholds	0.4419 vs. 0.3857 (primary); 0.4235 vs. 0.3763 (3-seed mean)	Adopted
Oversampling rare factors	Macro F1 0.4382 / 0.4297 (targets 20/40) vs. 0.4419	Rejected
Greedy span shrinking‡	0.5732 on one split; 0.5206 vs. 0.5246 mean	Rejected
CRF token tagging‡	0.5600 on one split; 0.5200 vs. 0.5246 mean	Rejected

‡As logged in the notebook's development notes; not re-executed in our verification run.

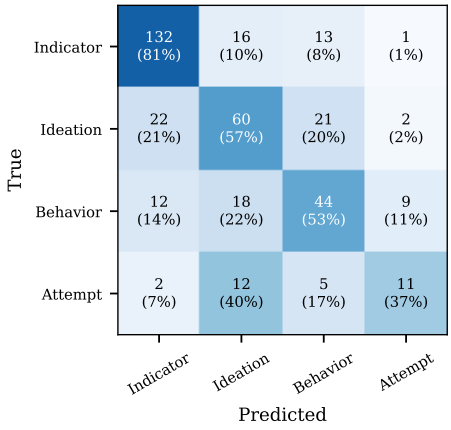


Fig. 6. Confusion matrix of the final risk model on the primary validation split (counts and row percentages).

TABLE VII
LOCAL PHRASE F1 BY TRUE RISK CLASS (PRIMARY SPLIT)

True class	Posts	Phrase F1
Indicator	161	0.882
Ideation	105	0.309
Behavior	83	0.327
Attempt	30	0.354
All	379	0.5599
Non-Indicator pooled	218	0.322

supported textual evidence; it is not clinical validation, and an extracted phrase may be correlated with, rather than causal for, the model's label.

Evidence quality is class-dependent. Table VII reports local Phrase F1 by true risk class on the 379 validation posts that have an evidence field. The overall 0.5599 is dominated by Indicator posts, where the reference is almost always "none" and an empty prediction is rewarded; on the other three classes the F1 is about one third (0.322 pooled). Applying the final pipeline's rule of suppressing evidence for posts predicted as Indicator changes the overall score only marginally (0.5608 vs. 0.5599).

X. ERROR ANALYSIS

Evidence errors. Among the 222 validation posts with a reference span, the extractor returns nothing for 53 and returns spans that match no reference for 65; the rest have at least one match. For the 157 posts annotated "none", 17 receive a spurious span. The main failure is granularity: predicted spans average 8.8 tokens against 4.4 for reference spans, i.e., the model often selects a whole clause where the annotator marked a short phrase, and the three-times length rule then rejects it. Further causes are posts with several evidence phrases (110 validation posts have more than one reference) where the system emits 0.88 spans per post on average against 1.03 reference spans, and evidence whose interpretation depends on context outside the candidate. Candidate boundaries come from punctuation and a fixed conjunction list, so they cannot isolate phrases inside an unpunctuated clause.

Factor errors. Performance tracks frequency (Table VIII). The common factors hopelessness, poor social support, coping strategy, and low self-esteem reach F1 of 0.67–0.73, while the six least frequent categories (at most 37 training posts) score 0.0–0.50 with 1–8 validation positives each, so these scores are not statistically stable. The model over-predicts: 3.79 factors per validation post against 2.78 annotated, and the discrepancy comes from long posts. For the longest quartile (more than 78 tokens), the model predicts 10.1 factors per post against 5.3 annotated, while for the remaining posts it predicts 1.7 against 2.0. On the prediction file, 31 of 502 posts receive more than ten factors. Long posts contain more matching vocabulary, and factors are correlated (e.g., hopelessness with coping strategy and emotion dysregulation), so one-vs-rest models, which score each factor independently, tend to fire together. According to the project documentation, a cap on the predicted factor count was tested and did not improve Macro F1, so it was left out; we document the behavior instead.

XI. RESPONSIBLE AI, ETHICS, AND LIMITATIONS

Research prototype. This system is a research prototype and is not a clinical diagnostic system. It should support trained human assessment rather than replace professional judgment, and we make no claim about its effect on real-world suicide prevention, which we have not measured.

TABLE VIII
PER-FACTOR RESULTS ON THE PRIMARY VALIDATION SPLIT (SUPPORT = POSITIVE POSTS; t_k = TUNED THRESHOLD)

Factor	Train	Val	t_k	F1	Factor	Train	Val	t_k	F1
hopelessness	595	150	.50	.713	interpersonal violence	59	24	.60	.537
emotion dysregulation	431	111	.45	.622	sense of responsibility	51	7	.25	.110
low self-esteem	373	102	.50	.669	low socio-economic status	42	12	.30	.444
poor social support	363	138	.45	.725	traumatic experience	41	23	.50	.500
coping strategy	331	112	.55	.703	meaning in life	37	8	.55	.154
suicide means (with access)	233	59	.45	.537	substance use	30	3	.45	.500
psychological capital	187	51	.60	.577	cognitive deficits	27	6	.50	.444
mental health issues	180	48	.50	.509	poor school performance	15	1	.20	.077
prior self-harm/suicidal thought/attempt	162	53	.50	.534	exposure to others' suicide	11	3	.20	.062
interpersonal difficulty	147	48	.55	.416	sexual orientation related issues	7	1	.20	.000
stressful life event	146	36	.60	.361	physical health/characteristic	71	7	.55	.267
dysfunctional family	90	28	.45	.686	social support	86	26	.45	.457

Risks. False negatives may leave a person in need unflagged; false positives may trigger unwarranted intervention or stigma. The validation F1 values (0.65 risk; 0.42 Attempt-class F1) show that both error types are frequent. Models trained on one community's language may transfer poorly across cultures, dialects, and platforms, and may encode demographic biases that we have not audited. The data are anonymized, but posts remain sensitive; any deployment would require consent-aware data governance, access control, and safeguards against misuse such as surveillance of individuals. Social-media text is also an incomplete signal of an individual's state.

Limitations as research boundaries. (i) The corpus is small (1,635 posts, 153 users), and splits resample the same users; (ii) several factors have fewer than 30 training posts; (iii) local Phrase F1 is an approximation and local validation need not match the official leaderboard; (iv) tuned thresholds and the evidence threshold were selected on the evaluation split, so single-split scores are optimistic; (v) final models were not refit on all labeled data; (vi) the Transformer pathway has no completed evaluation; and (vii) factor over-prediction on long posts remains.

XII. DISCUSSION

Why classical models work reasonably well. On a 1,635-post corpus, regularized linear models on sparse lexical features are competitive and cheap, with every weight inspectable. **Why character n-grams help:** they add robustness to spelling variation and partial lexical overlap at no modeling cost, and the gain was present on every seed, although modest ($+0.014$). They did not help for factors, where the topical content words are the signal and additional features likely add noise.

Why evidence and factors are harder. Evidence requires boundary decisions at phrase level against short, annotator-specific references, and the metric rewards an exact granularity; factors form a 24-way imbalanced problem with correlated labels and a long tail of tiny classes. **Why multi-seed validation matters:** the same baseline varies by 0.06 in Weighted F1 across splits, which is larger than most of the differences we examined, so single-split model selection would have adopted at least two spurious changes (Table VI).

What explainability adds, and its trade-offs. Verbatim evidence turns a label into something that can be checked and disputed, but its quality is uneven (Table VII) and it should not be read as a faithful account of the classifier's internal reasoning: the span scorer and the risk model are separate. Transformers could improve contextual understanding of severity but need more compute, are harder to inspect, and, given the validation-set size, would require the same multi-seed scrutiny; we make no claim about them.

XIII. CONCLUSION AND FUTURE WORK

We built and evaluated a user-grouped, multi-seed validated NLP pipeline that outputs a four-level risk label, verbatim evidence, and 24-way factor predictions. Risk classification is the strongest component (Weighted F1 0.6468 on the primary split, 0.6356 over five seeds), driven in part by word-plus-character features; evidence and factor identification remain the principal challenges, with evidence quality largely confined to the Indicator class. The main methodological finding is that single-split gains can mislead: several changes were rejected after multi-seed checks.

Future work includes completing and multi-seed evaluating Transformer models and larger pretrained models; nested or held-out threshold selection to remove the selection bias noted above; evidence extraction with learned span boundaries and richer context; handling of correlated and rare factors; calibration and uncertainty estimation; cross-domain validation; fairness analysis; and human-in-the-loop evaluation with clinicians. None of these has been implemented or evaluated here.

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